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FURTHER PALEOMAGNETIC RESULTS FROM THE JURASSIC TOPLEY INTRUSIVES IN THE STIKINIA SUBTERRANE OF BRITISH COLUMBIA

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Abstract. The 142 Ma old Jurassic Topley Intrusions are exposed in the middle of the Intermontane Belt or Stikinia subterrane of composite Terrane I of the Canadian Cordillera in central British Columbia. Thermal demagnetization at 560°C leads to the isolation of a single stable magnetic component directed at 324.2°, 54.1°, α_{95} = 8.6° in 17 of the 22 sites. The pole position of 238,2°W, 58.4°N, δ_p = 8.3°, δ_m = 11.8° for the component is discordant relative to the North American craton. This position supports previous work in suggesting that Stikinia has been translated northward by 14 ± 2° or about 1400 km between mid-Cretaceous and pre-Miocene time. It also suggests that the central portion of Stikinia has not been rotated about a vertical axis by a statistically significant amount whereas minor marginal blocks of the subterrane have been subjected to large and variable amounts of both clockwise and counterclockwise rotation.

Introduction

Recently Monger et al. (1982) subdivided composite Terrane I, essentially the Intermontane Belt of the Canadian Cordillera, into four subterranes including the Stikinia subterrane of northcentral British Columbia (Figure 1). Earlier Monger and Irving (1980) reported on paleomagnetic results from Stikinia. After alternating field (AF) demagnetization in the 30 to 60 mT range and thermal demagnetization in the 500 to 600°C range, they found that the Upper Triassic Takla Group volcanics, the Lower Jurassic Hazelton Group volcanics, and the mid-Cretaceous Axelgold layered gabbro intrusion from Stikinia all have remanence inclinations which indicate that they were formed at about 13° more southerly paleolatitudes than their present locations. Thus they concluded that Stikinia has moved northward by about 1300 km since mid-Cretaceous time. They also found large angular rotations in the remanence declinations about local vertical axes of 43° and 63° counterclockwise for the Takla sampling localities, of 41° and 103° counterclockwise and of 14° clockwise for the Hazelton sampling localities, and of 63° clockwise for the Axelgold sampling locality. Their results conflicted with my earlier data from the mid to Upper Jurassic Topley Intrusions (Symons, 1973a). After AF demagnetization in the 10 to 40 mT range, the Intrusions were interpreted to have a concordant pole position, and therefore to indicate neither translation or rotation for Stikinia. This conflict has created some problems in recent attempts to construct geotectonic models for the evolution of the Cordillera (e.g., Beck, 1980; Beck et al., 1981). During the past decade significant improvements have occurred in thermal demagnetization and component analysis techniques. Also many additional Jurassic poles have been added to the two available for the North American craton at the time of the 1973 report. Therefore, additional thermal demagnetization was done on the original collection to resolve the conflict in the data.

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Methods

The original collection of the Topley Intrusions (Symons, 1973a) contained 218 specimens cut from 109 cores representing 22 sites taken over an area of 200 km^2. The sites represent nine phases of the Intrusions giving K-Ar radiometric dates ranging from 163 to 112 Ma with seven of the phases tightly constrained between 133 and 143 Ma. Specimens were omitted that had been AF step demagnetized or glued together in the original study. Specimens with unstable or inhomogenous remanence were rejected also by requiring the within-specimen angular standard deviation to be less than 20° at a given measurement step (Harrison, 1980; Briden and Arthur, 1981; Stupavsky and Symons, 1982). Also a few specimens disintegrated in the oven and were excluded. This led to the acceptance of 136, 134 and 132 specimen vector directions at the three thermal cleaning temperatures of 200°C, 450°C and 560°C respectively. The vector directions for 30% of the accepted specimens had negative (up) inclina-

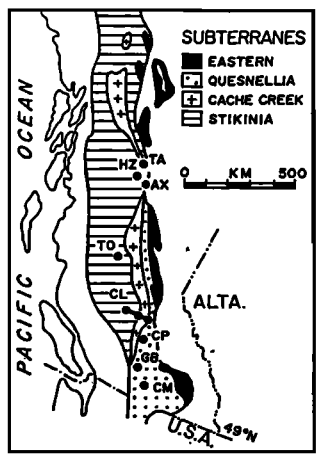


Fig. 1. Subterranes of Composite Terrane I in British Columbia and the southern Yukon after Monger et al. (1982) showing the locations of paleomagnetic collections for the Takla volcanics (TA), Hazelton volcanics (HZ), Axelgold intrusion (AX), Topley intrusions (TO), Cariboo lavas (CL), Cariboo plugs (CP), Guichon batholith (GB) and Copper Mountain plutons (CM).

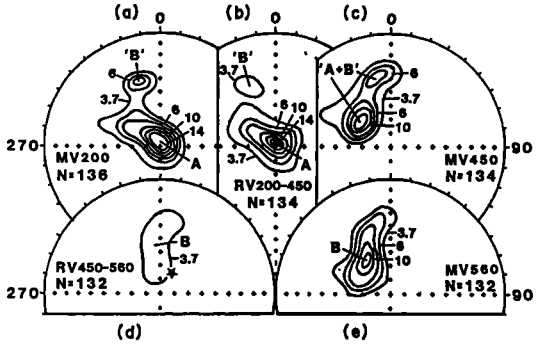


Fig. 2. Equal-area stereoplots showing the population density contours in 2% increments above the 95% confidence or 3.7% density contour for the: a) measured vectors (MV) after 200°C cleaning; b) vectors removed (VR) between 200 and 450°C; c) measured vectors at 450°C; d) vectors removed between 450 and 560°C; and e) measured vectors at 560°C. N is the number of vectors in the population. A and B are the isolated modes. The star in Fig. 2d indicates the present Earth's magnetic field direction.

tions, and these were rotated to their antiparallel or positive (down) position for analysis. Vector density distribution plots were generated for the measured residual vector at each temper-- step and for the removed vector between successive steps (Figure 2) following methods developed by Kamb (1959), Van Alstine (1980) and Stupavsky and Symons (1982).

Results

After thermal cleaning at 200°C (Figure 2a) the measured vectors (MV) form two nearly circular modes, A and B, of high statistical significance that are linked by an intervening girdle. A mode having a peak density of more than 3.7% (4.9% is statistically significant at the 95% (99%) confidence level for the 134 ± 2 specimen vectors involved in this study. The vertical A mode is attributed primarily to viscous remanence components acquired during storage since the original 1973 study and secondarily to residual underlying components. The lesser "B" mode is formed mainly by vectors from three phases of the Intrusions which Symons (1973a) found to carry the most stable remanence. The intervening girdle indicates the presence of hybrid "A + B" components.

The plot of the vector removed (VR) in the 200 and 450°C blocking temperature range (Figure 2b) shows the rapid preferential removal of the viscous A component. This is evident because the $VR_{200-450}$ A peak density (Figure 2b) is greater than the MV_{200} A peak density (Figure 2a). Also the directional coincidence of these two A modes indicates a clustering of linear decay segments in the equivalent Zijderveld plots (Zijderveld, 1967) for the individual specimens. Conversely the "B" component mode is being preferentially retained because the $VR_{200-450}$ peak density is substantially smaller than the MV_{200} peak density.

The MV_{450} plot after 450°C cleaning (Figure 2c) shows the shift in the "A" component direction towards the "B" direction along with a marked decrease in the "A" component density or statistical significance. This indicates that the "A" component is now a hybrid "A + B" component.

The $VR_{450-560}$ plot between 450°C and 560°C (Figure 2d) gives a single elongate mode that is coincident with the MV_{560} plot after 560°C cleaning (Figure 2e). This indicates that the softer low temperature A component has been mostly removed by 450°C and that the B component is undergoing unidirectional decay for most samples through the 450 to 560°C range. Also the highly significant peak density of 13% given for the B component mode in the MV_{560} plot relative to that of 6% in the $VR_{450-560}$ plot indicates that the B component has been well isolated by the 560°C cleaning in a highly significant number of specimens. Of the 132 specimens in the MV_{560} population, 72 fall within the 95% confidence limit. These come from 17 sites of which 11, 2 and 4 have normal, mixed and reversed polarities respectively. Thus only 5 of the original 22 sites did not retain a stable B component direction after 560°C cleaning. Given that most of the sampled Topley Intrusion phases fall in the quartz monzonite-granite-granodiorite compositional range, the presence of several unstable remanence sites is predictable. The elongate shape of the B mode appears to reflect secular variation in individual phases of the Topley Intrusions. For example, the two sites from the Francois phase give inclinations of 27° and 31°, and dominate the shallow inclination end of the mode. Conversely the two sites from the Glenannon and Stellako phases give inclinations of 72° and 68° respectively and dominate the steep inclination end of the mode.

The 17 site mean directions were calculated

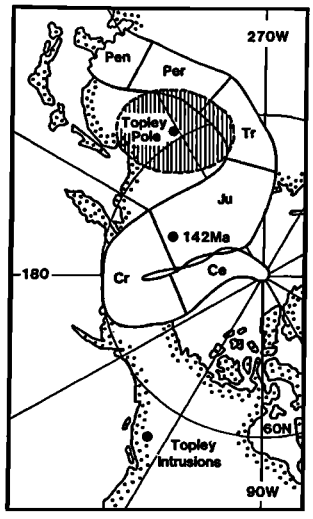


Fig. 3. Polar projection of the northern hemisphere with the apparent polar wander path for cratonic North America after Irving (1979) (Ce-Cenozoic, Cr-Cretaceous, Ju-Jurassic, Tr-Triassic, Per-Permian, and Pen-Pennsylvanian), the Topley intrusions sampling site, the 142 Ma cratonic reference pole and the pole position for the Topley intrusions circumscribed by its oval of 95% confidence.

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Table 1. Displacements Indicated by Paleomagnetism for Composite Terrane I

Unit	Terrane	Age	Paleolat. Translation	Clockwise Rotation	Reference
Cariboo plugs	1	Miocene	3° to S	0°	Symons (1969a)
Cariboo lavas	1	Miocene	6° to S	0°	Symons (1969b)
Axelgold intr.	Stikinia	l. Jur.-m. Cret.	13° to N	+63°	Monger & Irving (1980)
Topley intr.	Stikinia	m. - u. Jur.	18° to N	- 2°	this paper
Hazelton volc.	Stikinia	l. Jur.	14° to N	-103° to +14°	Monger & Irving (1980)
Takla volc.	Stikinia	u. Trias.	13° to N	-63° to -42°	Monger & Irving (1980)
Guichon bath.	Quesnellia	u. Trias.	13° to N	+43°	Symons (1983)
Copper Mtn. intr.	Quesnellia	u. Trias.	8° to S	+45°	Symons (1973b)

from the individual specimen vectors, and the site means combined to get the unit mean direction of 324.2°, 54.1°, α_{95} = 8.6° (declination, inclination, radius of cone of 95% confidence (Fisher, 1953)).

The unit mean direction gives a paleopole of 238.2°W, 58.4°N, δp = 8.3°, δ_m = 11.8° for the Topley Intrusions. This paleopole is shown in relation to the apparent polar wander (APW) path and 142 Ma reference pole for cratonic North America of Irving (1979) (Figure 3). The Topley pole indicates that central Stikinia has been translated poleward or northward by a statistically significant 18 ± 8° and rotated counterclockwise by a statistically nonsignificant 2° ± 12°.

Discussion

Monger and Irving (1980) and Monger et al. (1982) discuss the substantial stratigraphic, fault displacement and faunal assemblages evidence which supports the concept of northward translation for Stikinia. Miocene paleomagnetic data from the Cariboo lavas and Cariboo plugs (Table 1) of Terrane I indicate neither northward translation or rotation. Thus any motion of the terrane relative to the North American craton predates Miocene time.

The early Jurassic to mid-Cretaceous units all indicate from 13° to 18° of poleward or northward translation for the Stikinia subterrane, thereby confining this event to the mid-Cretaceous to pre-Miocene interval as Monger and Irving (1980) point out. However, the rotations are obviously at substantial variance. The Hazelton volcanics and Axelgold intrusion come from four locations with declination values that indicate rotations ranging from 103° counterclockwise to 63° clockwise. All four locations are close to the eastern margin of Stikinia. Conversely the Topley Intrusions are located in the central portion of Stikinia and do not indicate significant rotation. The simplest explanation is that Stikinia was translated northward by 14° ± 2° or about 1400 km with variable amounts of "ball bearing" type rotations (Beck and Noson, 1972) in small tectonic blocks along Stikinia's margin as it slid differentially past the Omineca Crystalline Belt (Monger et al., 1982) to the east. Similar variable rotations are found elsewhere in the Cordillera along the margins of terrane-sized tectonic units (Simpson and Cox, 1977; Kammerling and Luyendyk, 1979). In contrast the central portion of Stikinia where the Topley Intrusions are exposed behaved as a coherent block during the translation event.

The late Triassic paleomagnetic data from Terrane I provides further limitations. The Takla volcanics were sampled at two locations along the eastern margin of Stikinia (Monger and Irving, 1980). They similarly show 13° of northward translation and large variable counterclockwise rotations. The Guichon Batholith outcrops in the Quesnellia subterrane of composite Terrane I (Monger et al., 1982). Its paleopole (Symons, 1983) also indicates 13° ± 6° of northward translation. This suggests that Quesnellia has maintained the same paleolatitudinal position relative to Stikinia since Guichon intrusion at 198 ± 8 Ma ago. The 43° ± 7° clockwise rotation of the Guichon paleopole postdates the assembling of Terrane I but could reflect either of two events. One is the subduction event associated with the formation of the Cache Creek subterrane which lies between Quesnellia and Stikinia, and the other is a differential terrane rotation event associated with the northward translation. The clockwise rotation of Quesnellia is consistent with the 45° clockwise rotation found for the Copper Mountain plutons paleopole (Symons, 1973b), however, this paleopole has an inconsistent paleolatitude which likely stems from incomplete AF demagnetization using 10 to 40 mT fields in the original study. Further work is planned for these plutons to examine the inconsistency.

In summary, the revised pole for the Topley Intrusions resolves the paleolatitude conflict created by the original data, and thereby simplifies the construction of geotectonic models for Cordilleran evolution. It also provides evidence of the need to sample rock units that are internal to a terrane in order to avoid localized rotations of minor blocks at the margins of a terrane.

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